JANANI J,

192512155.

EXPERIMENT – 4

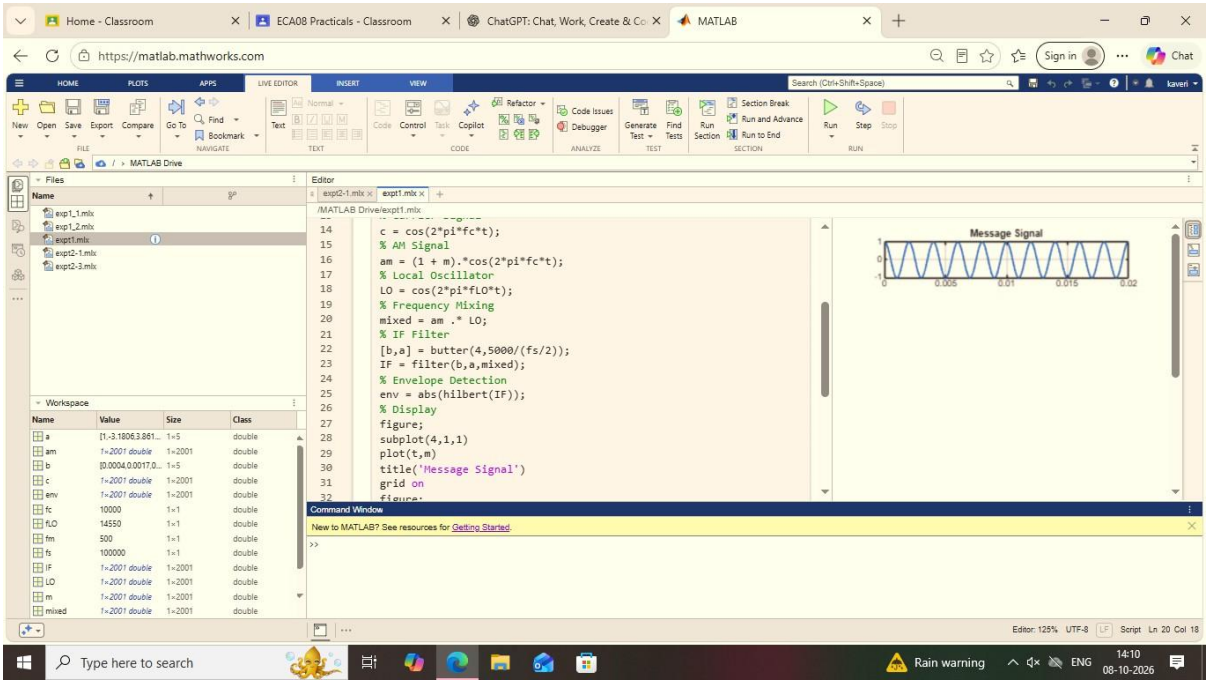
MATLAB SIMULATION OF SUPERHETERODYNE RECEIVER FOR AM COMMUNICATION SYSTEM WITH FREQUENCY MIXING AND FILTERING

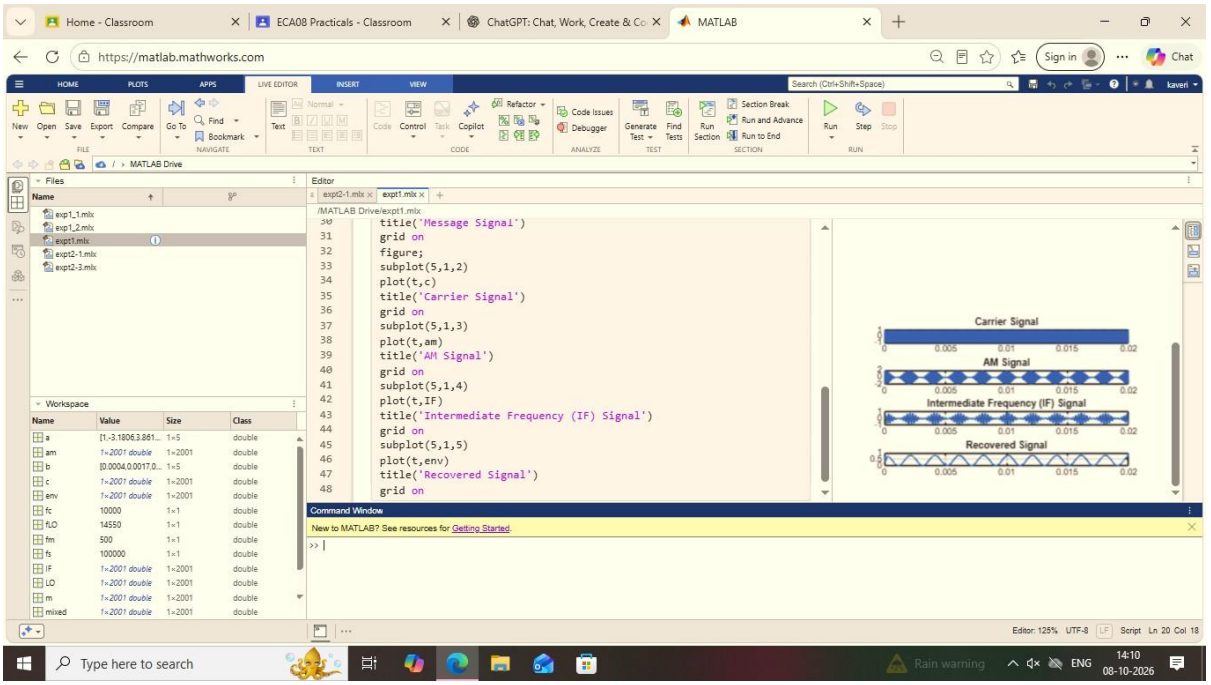
PROGRAM:

```
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 10000; % Carrier frequency
fLO = 14550; % Local Oscillator (IF = 4550 Hz)
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t);
% Frequency Mixing
mixed = am .* LO;
% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);
% Envelope Detection
env = abs(hilbert(IF));
```

```
% Display
figure;
subplot(4,1,1)
plot(t,m)
title('Message Signal')
grid on
figure;
subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on
subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on
subplot(5,1,4)
plot(t,IF)
title('Intermediate Frequency (IF) Signal')
grid on
subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
grid on
```

OUTPUT:


Message Signal
Command Window
New to MATLAB? See resources for Getting Started.


Carrier Signal
AM Signal
Intermediate Frequency (IF) Signal
Recovered Signal
Command Window
New to MATLAB? See resources for Getting Started.